ShopEase REST API Specification

Version 1.0 | Base URL: `https://api.shopease.example.com/v1` | Format: JSON | Auth: JWT Bearer

This sample document describes the endpoints of a fictional e-commerce API. Each endpoint section lists the HTTP method, headers, path/query parameters, request fields, response fields with data types, status codes and a sample response.

1. Endpoint Summary

ID	Endpoint Name	Method	Path	Auth	Success Code
API-01	User Login	POST	`/auth/login`	None	200
API-02	Create User	POST	`/users`	Bearer token (admin)	201
API-03	Get User by ID	GET	`/users/{userId}`	Bearer token	200
API-04	Update User	PUT	`/users/{userId}`	Bearer token	200
API-05	Delete User	DELETE	`/users/{userId}`	Bearer token (admin)	200
API-06	List Products	GET	`/products`	None	200
API-07	Create Order	POST	`/orders`	Bearer token	201
API-08	Get Order Details	GET	`/orders/{orderId}`	Bearer token	200
API-09	Update Order Status	PATCH	`/orders/{orderId}/status`	Bearer token (admin)	200

2. Common Request Headers

Header	Value / Format	Required	Description
Content-Type	application/json	Yes	Media type of the request body
Accept	application/json	Yes	Expected response format
Authorization	Bearer <access_token>	Yes	JWT token from /auth/login
X-Request-ID	uuid string	No	Correlation ID for tracing

3. Common Error Response

Field	Data Type	Description
timestamp	string (date-time)	Time of the error
status	integer	HTTP status code
error	string	Short error code, e.g. VALIDATION_ERROR
message	string	Human-readable message
path	string	Request path

```
{
  "timestamp": "2026-10-23T10:20:00Z",
  "status": 400,
  "error": "VALIDATION_ERROR",
  "message": "email must be a valid email address",
  "path": "/v1/users"
}
```

4. Endpoint Details

API-01 User Login

Attribute	Value
Method	**POST**
Endpoint	`/auth/login`
Full URL	`https://api.shopease.example.com/v1/auth/login`
Description	Authenticates a user and returns an access token.
Authentication	None

Request Headers

Header	Value / Format	Required	Description
Content-Type	application/json	Yes	Media type of the request body
Accept	application/json	Yes	Expected response format
X-Request-ID	uuid string	No	Correlation ID for tracing

Request Body Fields

Field	Data Type	Required	Description
email	string	Yes	Registered email address
password	string	Yes	Account password (min 8 chars)

Response Fields

Field	Data Type	Description
accessToken	string	JWT access token
tokenType	string	Always 'Bearer'
expiresIn	integer	Token validity in seconds
user.id	integer	Unique user ID
user.name	string	Full name of the user

Status Codes

Code	Meaning
200	Login successful
400	Missing email or password
401	Invalid credentials

Sample Response (200)

```
{
  "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "tokenType": "Bearer",
  "expiresIn": 3600,
  "user": {
    "id": 1024,
    "name": "Priya Raman"
  }
}
```

API-02 Create User

Attribute	Value
Method	**POST**
Endpoint	`/users`
Full URL	`https://api.shopease.example.com/v1/users`
Description	Registers a new customer account.
Authentication	Bearer token (admin)

Request Headers

Header	Value / Format	Required	Description
Content-Type	application/json	Yes	Media type of the request body
Accept	application/json	Yes	Expected response format
Authorization	Bearer <access_token>	Yes	JWT token from /auth/login
X-Request-ID	uuid string	No	Correlation ID for tracing

Request Body Fields

Field	Data Type	Required	Description
name	string	Yes	Full name (2-100 chars)
email	string	Yes	Unique email address
phone	string	No	Phone number in E.164 format
role	string (enum)	Yes	CUSTOMER | ADMIN
isActive	boolean	No	Defaults to true

Response Fields

Field	Data Type	Description
id	integer	Generated user ID
name	string	Full name
email	string	Email address
role	string	Assigned role
createdAt	string (date-time)	ISO 8601 creation timestamp

Status Codes

Code	Meaning
201	User created
400	Validation error
409	Email already exists

Sample Response (201)

```
{
  "id": 1025,
  "name": "Arjun Mehta",
  "email": "arjun.mehta@example.com",
  "role": "CUSTOMER",
  "createdAt": "2026-10-23T10:15:30Z"
}
```

API-03 Get User by ID

Attribute	Value
Method	**GET**
Endpoint	`/users/{userId}`
Full URL	`https://api.shopease.example.com/v1/users/{userId}`
Description	Fetches the details of a single user.
Authentication	Bearer token

Request Headers

Header	Value / Format	Required	Description
Accept	application/json	Yes	Expected response format
Authorization	Bearer <access_token>	Yes	JWT token from /auth/login
X-Request-ID	uuid string	No	Correlation ID for tracing

Path / Query Parameters

Parameter	In	Data Type	Required	Description
userId	path	integer	Yes	ID of the user

Request Body Fields

No request body.

Response Fields

Field	Data Type	Description
id	integer	User ID
name	string	Full name
email	string	Email address
phone	string	Phone number
role	string	CUSTOMER | ADMIN
isActive	boolean	Account status
address	object	Default shipping address
address.city	string	City
address.pincode	string	Postal code

Status Codes

Code	Meaning
200	User found
401	Unauthorized
404	User not found

Sample Response (200)

```
{
  "id": 1024,
  "name": "Priya Raman",
  "email": "priya.raman@example.com",
  "phone": "+919876543210",
  "role": "CUSTOMER",
  "isActive": true,
  "address": {
    "city": "Chennai",
    "pincode": "600042"
  }
}
```

API-04 Update User

Attribute	Value
Method	**PUT**
Endpoint	`/users/{userId}`
Full URL	`https://api.shopease.example.com/v1/users/{userId}`
Description	Replaces the editable details of an existing user.
Authentication	Bearer token

Request Headers

Header	Value / Format	Required	Description
Content-Type	application/json	Yes	Media type of the request body
Accept	application/json	Yes	Expected response format
Authorization	Bearer <access_token>	Yes	JWT token from /auth/login
X-Request-ID	uuid string	No	Correlation ID for tracing

Path / Query Parameters

Parameter	In	Data Type	Required	Description
userId	path	integer	Yes	ID of the user

Request Body Fields

Field	Data Type	Required	Description
name	string	Yes	Full name
phone	string	No	Phone number
address.city	string	No	City
address.pincode	string	No	6-digit postal code

Response Fields

Field	Data Type	Description
id	integer	User ID
name	string	Updated name
updatedAt	string (date-time)	Last update timestamp

Status Codes

Code	Meaning
200	User updated
400	Validation error
404	User not found

Sample Response (200)

```
{
  "id": 1024,
  "name": "Priya R. Raman",
  "updatedAt": "2026-10-23T11:02:10Z"
}
```

API-05 Delete User

Attribute	Value
Method	**DELETE**
Endpoint	`/users/{userId}`
Full URL	`https://api.shopease.example.com/v1/users/{userId}`
Description	Soft-deletes a user account.
Authentication	Bearer token (admin)

Request Headers

Header	Value / Format	Required	Description
Accept	application/json	Yes	Expected response format
Authorization	Bearer <access_token>	Yes	JWT token from /auth/login
X-Request-ID	uuid string	No	Correlation ID for tracing

Path / Query Parameters

Parameter	In	Data Type	Required	Description
userId	path	integer	Yes	ID of the user

Request Body Fields

No request body.

Response Fields

Field	Data Type	Description
message	string	Confirmation message
deletedId	integer	ID of the deleted user

Status Codes

Code	Meaning
200	User deleted
403	Forbidden (not admin)
404	User not found

Sample Response (200)

```
{
  "message": "User deleted successfully",
  "deletedId": 1025
}
```

API-06 List Products

Attribute	Value
Method	**GET**
Endpoint	`/products`
Full URL	`https://api.shopease.example.com/v1/products`
Description	Returns a paginated list of products with optional filters.
Authentication	None

Request Headers

Header	Value / Format	Required	Description
Accept	application/json	Yes	Expected response format
X-Request-ID	uuid string	No	Correlation ID for tracing

Path / Query Parameters

Parameter	In	Data Type	Required	Description
category	query	string	No	Filter by category
minPrice	query	number	No	Minimum price
maxPrice	query	number	No	Maximum price
page	query	integer	No	Page number (default 1)
size	query	integer	No	Items per page (default 20, max 100)

Request Body Fields

No request body.

Response Fields

Field	Data Type	Description
page	integer	Current page
size	integer	Page size
totalItems	integer	Total matching products
items	array[object]	List of products
items[].productId	string	Product SKU
items[].name	string	Product name
items[].price	number (decimal)	Unit price in INR
items[].inStock	boolean	Availability flag

Status Codes

Code	Meaning
200	Products returned
400	Invalid filter value

Sample Response (200)

```
{
  "page": 1,
  "size": 2,
  "totalItems": 48,
  "items": [
    {
      "productId": "SKU-1001",
      "name": "Wireless Mouse",
      "price": 799.0,
      "inStock": true
    },
    {
      "productId": "SKU-1002",
      "name": "USB-C Hub",
      "price": 1899.5,
      "inStock": false
    }
  ]
}
```

API-07 Create Order

Attribute	Value
Method	**POST**
Endpoint	`/orders`
Full URL	`https://api.shopease.example.com/v1/orders`
Description	Places a new order for the authenticated user.
Authentication	Bearer token

Request Headers

Header	Value / Format	Required	Description
Content-Type	application/json	Yes	Media type of the request body
Accept	application/json	Yes	Expected response format
Authorization	Bearer <access_token>	Yes	JWT token from /auth/login
X-Request-ID	uuid string	No	Correlation ID for tracing
Idempotency-Key	uuid string	Yes	Prevents duplicate order creation

Request Body Fields

Field	Data Type	Required	Description
items	array[object]	Yes	Products to order (min 1)
items[].productId	string	Yes	Product SKU
items[].quantity	integer	Yes	Quantity (1-10)
paymentMode	string (enum)	Yes	UPI | CARD | COD
couponCode	string	No	Discount coupon

Response Fields

Field	Data Type	Description
orderId	string	Generated order ID
status	string	Initial status: PLACED
totalAmount	number (decimal)	Payable amount after discount
currency	string	ISO currency code
estimatedDelivery	string (date)	Expected delivery date

Status Codes

Code	Meaning
201	Order created
400	Invalid items or quantity
409	Product out of stock
422	Invalid coupon

Sample Response (201)

```
{
  "orderId": "ORD-20261023-5521",
  "status": "PLACED",
  "totalAmount": 2498.5,
  "currency": "INR",
  "estimatedDelivery": "2026-10-28"
}
```

API-08 Get Order Details

Attribute	Value
Method	**GET**
Endpoint	`/orders/{orderId}`
Full URL	`https://api.shopease.example.com/v1/orders/{orderId}`
Description	Fetches an order with its line items.
Authentication	Bearer token

Request Headers

Header	Value / Format	Required	Description
Accept	application/json	Yes	Expected response format
Authorization	Bearer <access_token>	Yes	JWT token from /auth/login
X-Request-ID	uuid string	No	Correlation ID for tracing

Path / Query Parameters

Parameter	In	Data Type	Required	Description
orderId	path	string	Yes	Order ID

Request Body Fields

No request body.

Response Fields

Field	Data Type	Description
orderId	string	Order ID
userId	integer	Owner of the order
status	string (enum)	PLACED | SHIPPED | DELIVERED | CANCELLED
items	array[object]	Line items
items[].productId	string	Product SKU
items[].quantity	integer	Ordered quantity
items[].lineTotal	number (decimal)	Price x quantity
totalAmount	number (decimal)	Order total
createdAt	string (date-time)	Order timestamp

Status Codes

Code	Meaning
200	Order found
403	Order belongs to another user
404	Order not found

Sample Response (200)

```
{
  "orderId": "ORD-20261023-5521",
  "userId": 1024,
  "status": "PLACED",
  "items": [
    {
      "productId": "SKU-1001",
      "quantity": 1,
      "lineTotal": 799.0
    },
    {
      "productId": "SKU-1003",
      "quantity": 1,
      "lineTotal": 1699.5
    }
  ],
  "totalAmount": 2498.5,
  "createdAt": "2026-10-23T12:40:05Z"
}
```

API-09 Update Order Status

Attribute	Value
Method	**PATCH**
Endpoint	`/orders/{orderId}/status`
Full URL	`https://api.shopease.example.com/v1/orders/{orderId}/status`
Description	Moves an order to a new status (admin only).
Authentication	Bearer token (admin)

Request Headers

Header	Value / Format	Required	Description
Content-Type	application/json	Yes	Media type of the request body
Accept	application/json	Yes	Expected response format
Authorization	Bearer <access_token>	Yes	JWT token from /auth/login
X-Request-ID	uuid string	No	Correlation ID for tracing

Path / Query Parameters

Parameter	In	Data Type	Required	Description
orderId	path	string	Yes	Order ID

Request Body Fields

Field	Data Type	Required	Description
status	string (enum)	Yes	SHIPPED | DELIVERED | CANCELLED
remarks	string	No	Reason or note (max 250 chars)

Response Fields

Field	Data Type	Description
orderId	string	Order ID
previousStatus	string	Status before update
currentStatus	string	Status after update
updatedAt	string (date-time)	Update timestamp

Status Codes

Code	Meaning
200	Status updated
400	Invalid status transition
403	Forbidden (not admin)
404	Order not found

Sample Response (200)

```
{
  "orderId": "ORD-20261023-5521",
  "previousStatus": "PLACED",
  "currentStatus": "SHIPPED",
  "updatedAt": "2026-10-24T09:00:00Z"
}
```